Table 2: Leading quantum algorithms

Algorithm	What it does	Speedup/Type	Typical use case
Shor's algorithm	Factors large integers and solves discrete logarithms in polynomial time	Exponential (over best-known classical)	Breaking RSA/ ECC encryption; the primary driver of post-quantum cryptography migration
Grover's algorithm	Searches an unsorted database of N items	Quadratic ($\sqrt{N}$ vs N classically)	Unstructured search, database lookup, brute-force cryptanalysis of symmetric keys
Quantum Fourier Transform (QFT)	Quantum analogue of the discrete Fourier transform, converted into a sequence of quantum gates	Exponential over classical FFT for certain uses	Core subroutine inside Shor's algorithm and quantum phase estimation
Variational Quantum Eigensolver (VQE)	Hybrid quantum-classical algorithm that estimates the lowest eigenvalue (ground-state energy) of a molecule or material	Problem-dependent; near-term/NISQ-friendly	Molecular simulation, drug discovery, catalyst and battery-material design
Quantum Approximate Optimization Algorithm (QAOA)	Hybrid algorithm that approximates optimal solutions to combinatorial optimisation problems	Heuristic speedup on specific problem structures	Portfolio optimisation, route planning, scheduling, max-cut style problems
HHL (Harrow-Hassidim-Lloyd) algorithm	Solves systems of linear equations by encoding the solution in quantum amplitudes	Exponential under specific sparsity/ conditioning assumptions	Foundational subroutine for quantum machine learning and quantum-enhanced linear algebra
Quantum Phase Estimation (QPE)	Estimates the eigenvalue (phase) associated with a unitary operator's eigenvector	Exponential precision scaling	Building block for Shor's algorithm, VQE refinement, and chemistry simulation
Quantum Annealing	Uses adiabatic evolution to settle a system into its lowest-energy configuration	Heuristic; no proven general speedup	Combinatorial optimisation on annealing hardware (D-Wave)
Quantum Machine Learning (QSVM, QNN)	Quantum-native variants of support vector machines and neural networks using quantum feature spaces	Problem-dependent; theoretical speedups for specific data structures	Pattern classification, anomaly detection, high-dimensional data analysis

Rigetti, D-Wave, and Oxford Quantum Circuits hardware.

Nvidia: In March 2025, Nvidia launched the Nvidia Accelerated Quantum Research Center (NVAQC) in Boston. This centre is all about developing new quantum computing architectures and algorithms that can work together with their powerful AI supercomputers.

IonQ: IonQ has developed quantum computers based on trapped ion technology.

Alibaba: Alibaba has designed an experimental quantum computer which is available online. In March 2018, it had launched its 'superconducting quantum computing cloud' to provide access to an 11-qubit quantum computer. Developed along with the Chinese Academy of Sciences, it allows users to run quantum programs and download the results.

Quantinuum: Formed from the merger of Cambridge Quantum and Honeywell Quantum Solutions, Quantinuum builds high-fidelity trapped-ion systems (the H-series and Helios). It filed for a US IPO in January 2026, alongside Microsoft, on the strength of its logical-qubit demonstrations.

QuEra Computing: Spun out of Harvard and MIT research, QuEra uses neutral atoms held in optical tweezer arrays as qubits, and has demonstrated some of the highest logical qubit counts in the industry on its Aquila hardware.

Other leading quantum computing vendors are Xanadu, Pasqal, Riverlane, Honeywell International, and Huawei Technologies Co Ltd.

The implications of true quantum computing at scale are staggering and of extraordinary impact to society. Quantum algorithms are applied for:

- Optimisation problems such as scheduling and route planning to